University of Florida, Department of Mathematics

Analysis First-Year Exam, May 2025, Part 1

Answer FOUR questions in detail. State carefully any results used without proof.

1. Let A be a subset of the metric space M. Decide (with proof or counterexample) whether the following statements are equivalent:

(i) A is open;

(ii) for each $X \subseteq M$, if $A \cap X = \varnothing$ then $A \cap \overline{X} = \varnothing$.

2. The function $f : X \to Y$ between metric spaces is distance-preserving. Must $f(X)$ be closed in Y in each of the following cases? If the answer is Yes, give a proof; if the answer is No, give a counterexample.

(i) X is compact.

(ii) X is complete.

3. Let $f : X \to Y$ be a continuous bijection, $g : Y \to X$ its inverse. Must g be continuous in each of the following cases? Proof or counterexample, as appropriate.

(i) X is compact.

(ii) Y is compact.

4. The function $f : \mathbb{R} \to \mathbb{R}$ is continuous and satisfies

$$\liminf f(n) < 0 < \limsup f(n)$$

where n runs through the natural numbers. Prove that each of $f^{-1}(-\infty, 0)$ and $f^{-1}(0, \infty)$ is infinite.

5. Let $f : \mathbb{R} \to \mathbb{R}$ be such that

$$|f(x) - f(y)| \leqslant |x - y|^a$$

for some $a > 1$ and all $x, y \in \mathbb{R}$. Deduce as much as possible about f.